Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 30)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Which of the following oil spill treatments is most likely to kill organisms living deep in beach sediments that were not affected by the oil? Tick one box.

[1 mark]

A Absorbent mops ☐

B Steam washing ☐

C Inflatable booms ☐

D Polymerising agents ☐

0 1 . 2 Explain how polymerising agents make an oil spill easier to clean up.

[2 marks]

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0 1 . 3 Explain why deoxygenation is a more common problem in water than in air.

[2 marks]

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0 2 . 1 Describe polymetallic (manganese) nodules and explain why they are not yet exploited on a large scale.

[3 marks]

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0 2 . 2 Explain the environmental impacts of mining polymetallic nodules.

[3 marks]

0 2 . 3 Explain why rare earth metals are a supply concern and describe one new extraction technique.

[4 marks]

10

0 3 . 1 Explain why planting a new forest only sequesters carbon for a limited period, and how the storage can be extended.

[3 marks]

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0 3 . 2 Compare pre-combustion and post-combustion methods of carbon capture.

[4 marks]

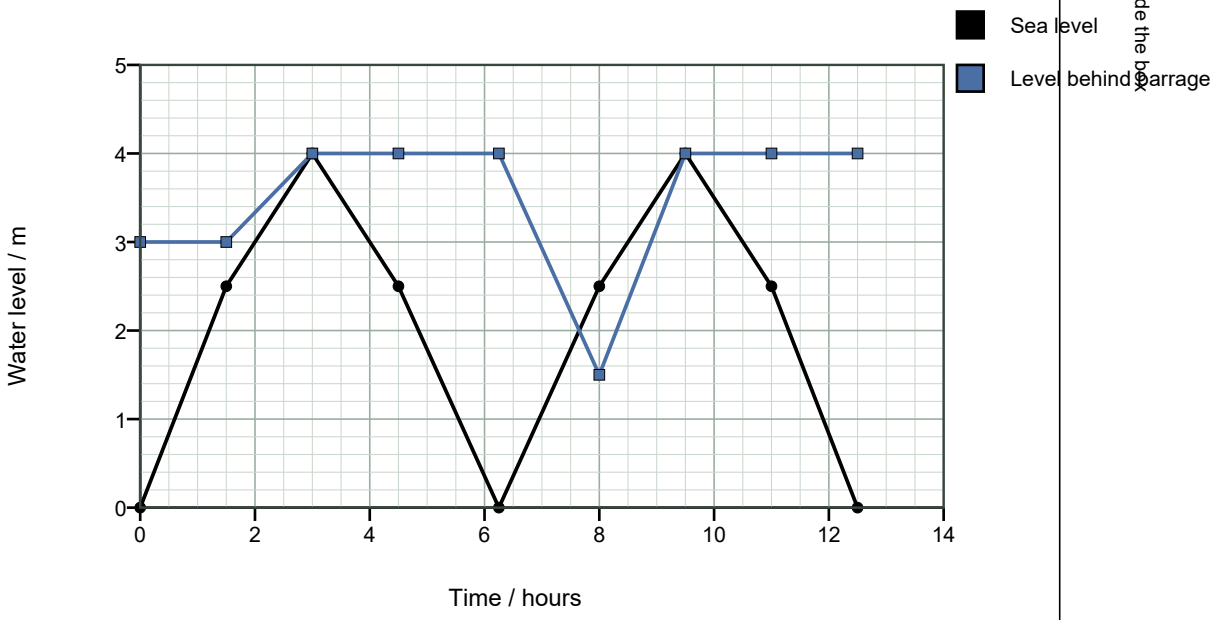
0 3 . 3 Describe **two** ways captured CO_2 can be stored and give **one** concern about long-term storage.

[3 marks]

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0 4 Figure 1 shows the water level inside and outside a tidal barrage over one 12.5-hour tidal cycle and the periods when electricity is generated.

Figure 1



0 4 . 1 Use Figure 1 to explain when electricity is generated by the barrage.

[2 marks]

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0 4 . 2 Explain why tidal power is intermittent but predictable.

[2 marks]

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0 4 . 3 Explain the difference between spring and neap tides and why this affects output.

[2 marks]

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0 4 . 4 Explain how a tidal lagoon can be used to generate electricity when there is no tidal flow.

[2 marks]

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0 4 . 5 Give **two** coastal features that make a site suitable for tidal power.

[2 marks]

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2 .....

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0 5 . 1 Which of the following is the main reason that lead-based paint is a greater risk to children than to adults? Tick one box.

[1 mark]

A Children inhale more lead vapour ☐

B Children pick up dust / flakes and suck their fingers ☐

C Lead is more soluble in children's blood ☐

D Children spend more time near roads ☐

0 5 . 2 Explain how tri-butyl tin (TBT) in anti-fouling paint caused pollution even though only small amounts dissolved into the sea.

[3 marks]

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0 5 . 3 Iron is not toxic, yet iron-rich drainage from mine spoil heaps kills fish. Explain why, and describe how it is controlled.

[3 marks]

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0 5 . 4 Give **two** uses of lead that continue because they do not cause significant lead release into the environment.

[2 marks]

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0 5 . 5 Explain why heavy metals are described as neurotoxins.

[1 mark]

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0 6 . 1 Define the term **persistence** as applied to a pollutant.

[1 mark]

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0 6 . 2 Explain how the persistence of a pesticide is affected by **two** named environmental factors.

[4 marks]

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0 7 Table 1 shows the running mean of moth catches as a student increased the number of nights of light-trapping at one site.

Table 1

Number of nights	**1**	**2**	**4**	**6**	**8**	**10**	**12**	**14**
Running mean catch per night	31	18	24	21	22	22	21	22

0 7 . 1 Use Table 1 to suggest the minimum number of nights needed to obtain a reliable mean. Explain your answer.

[2 marks]

0 7 . 2 Explain why moth surveys must be carried out at standardised times and weather conditions.

[3 marks]

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0 7 . 3 Distinguish between **accuracy** and **precision** of measurements.

[2 marks]

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0 7 . 4 Explain the importance of a null hypothesis and standardised techniques in a scientific investigation.

[3 marks]

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0 8 Figure 1 shows a simplified energy budget for the Earth and its atmosphere. Values are percentages of incoming solar radiation.

Figure 1

Energy flow	Percentage of incoming solar radiation / %
Reflected by clouds, dust and atmosphere	25
Reflected by the Earth's surface	5
Absorbed by the atmosphere	20
Absorbed by the Earth's surface	50
Infrared emitted by the surface and absorbed by greenhouse gases	95
Infrared re-emitted by the atmosphere back to the surface	85
Infrared emitted to space	70

0 8 . 1 Use Figure 1 to calculate the planetary albedo of the Earth.

[1 mark]

Answer ____ %

0 8 . 2 Describe how each of the following wavelengths of electromagnetic radiation behaves in the atmosphere.

[6 marks]

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Ultraviolet (UV) .....

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Visible light .....

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Infrared (IR) .....

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0 8 . 3 Explain why the amount of infrared re-emitted to the surface (85%) is larger than the amount emitted to space (70%).

[2 marks]

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0 8 . 4 Explain the term **thermal stratification** of the atmosphere.

[1 mark]

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0 9 . 1 Explain how occupancy sensors and programmable thermostats reduce energy waste.

[2 marks]

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0 9 . 2 Explain why LED lights use much less electricity than the filament and CFL lamps they replaced, and give **two** other examples of low-energy appliance design.

[3 marks]

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0 9 . 3 Describe **three** changes in human behaviour that reduce domestic energy use.

[3 marks]

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0 9 . 4 Explain why heating water on demand saves energy compared with storing hot water.

[2 marks]

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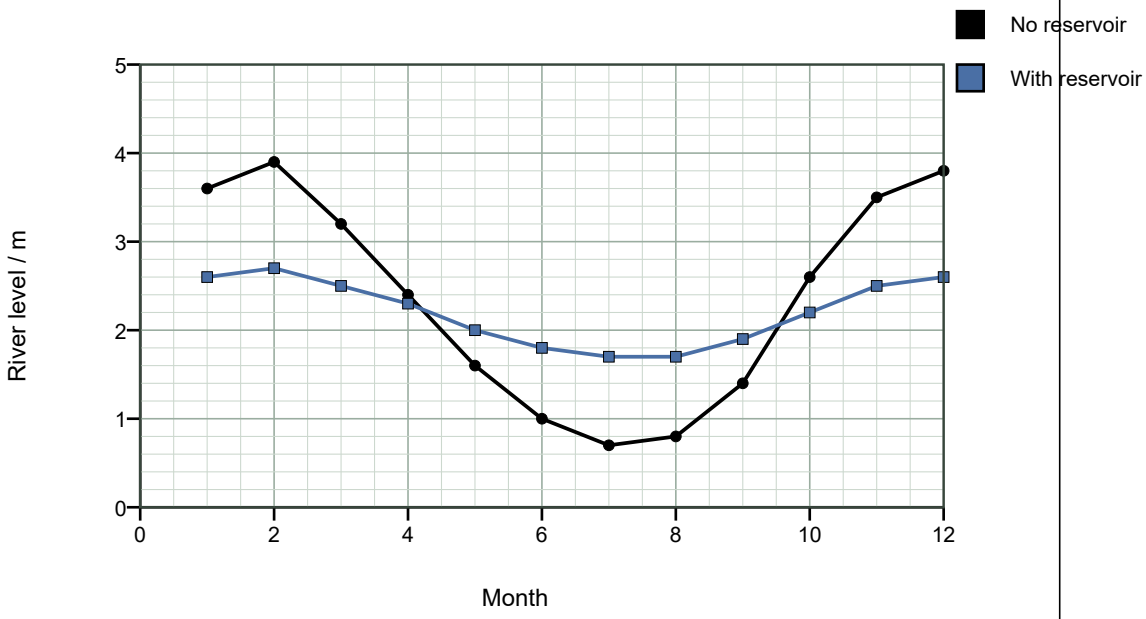
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1 0 Figure 1 shows the river level below a reservoir site during one year, before and after a river-regulation reservoir was built.

Figure 1



1 0 . 1 Use Figure 1 to calculate the reduction in the annual range of river level produced by the reservoir.

[2 marks]

Answer ..... m

1 0 . 2 Explain how the reservoir produces the changes shown in Figure 1.

[2 marks]

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1 0 . 3 Explain why human activities in the catchment may have increased the extremes in river flow before the reservoir was built.

[2 marks]

1 0 . 4 Discuss the factors that should be considered when selecting a site for a new reservoir, and evaluate the alternative strategies for increasing the sustainability of water supplies.

[9 marks]

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Extra space

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Write an essay on one of the following topics.

1 1 . 1 Discuss how knowledge of the properties of pollutants can be used to reduce the severity of their impact on the environment.

[25 marks]

OR

1 1 . 2 Soil is often described as a non-renewable resource. Discuss the causes and consequences of soil degradation and evaluate the strategies that can be used to manage soils sustainably.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS